

DISPERSIVE CHARACTERISTICS OF THE FIRST THREE RAYLEIGH MODES FOR A SINGLE SURFACE LAYER

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ABSTRACT

The dispersive characteristics of a single elastic layer overlying an elastic half-space are examined in detail for the fundamental and the first and second higher modes of Rayleigh waves. Phase velocity, group velocity, and the ratio of horizontal to vertical surface displacement are computed as functions of dimensionless quantities proportional to period and wave number.

The significant range for the independent variable, B_1T/H , proves to be largely independent of the parameters of the structure. The range is 1 to 20 for the fundamental, 0.3 to cutoff for the first higher mode, and 0.2 to cutoff for the second higher mode.

The most important parameter of the structure for Rayleigh wave dispersion is the shear velocity ratio. Variations in the Poisson's ratio in the surface layer and the density contrast may produce substantial effects. Poisson's ratio in the half-space is of least significance.

The dependence on model parameters of the long-period cutoff for the higher modes is determined.

Specific results are given for the following geophysical examples: continental crust, continental ice cap, sedimentary basin, alluvial overburden, and laboratory seismic models.

INTRODUCTION

The purpose of this work is to exhibit in a complete manner the characteristics of Rayleigh wave dispersion in a uniform elastic layer overlying a uniform elastic half space. In particular, we wish to show by a numerical analysis how the dispersion depends upon each of the four dimensionless parameters characterizing the structure.

The computed dependent variables are phase velocity, group velocity, and the ratio of horizontal to vertical surface displacement. Precision in the values of group velocity is achieved by carrying out differentiations analytically rather than numerically.

For the graphical presentation, a logarithmic scale is used for the independent variable, B_1T/H (see below for nomenclature). This makes the shape and size of the dispersion curve independent of the layer thickness; any desired thickness value can be obtained by horizontal translation of the curve.

Results are given for the first and second higher Rayleigh modes as well as the fundamental. Few observational data are available on the higher Rayleigh-type modes, but several recent papers indicate increasing interest in the subject (e.g., Oliver and Ewing, 1957; Oliver and Ewing, 1958; Oliver, Dorman, and Sutton, 1959; Levshin, 1962; Crampin, 1964; Kovach and Anderson, 1964). An extensive work on the Rayleigh wave higher modes is that of Alexander (1963) who devised a

method for discriminating between modes, and used the results to reduce the ambiguities inherent in dispersion interpretation.

Previous work which is directed toward a systematic examination of the dispersive properties of simple models for Rayleigh waves has been limited. Early papers of Sezawa and Kanai (1935a, 1935b) contain scattered examples. Giese (1957) and Tazine (1959) dealt with the limiting case of a rigid substratum. Somewhat restricted tabulations of dispersion curves for the fundamental and first higher mode of Rayleigh waves were computed for eleven related two-layered models by Bolt and Butcher (1960).

NOMENCLATURE AND CHOICE OF PARAMETERS

The structural model, shown in Figure 1, consists of two uniform solid perfectly elastic media: a layer of thickness H , overlying a half space. The following symbols

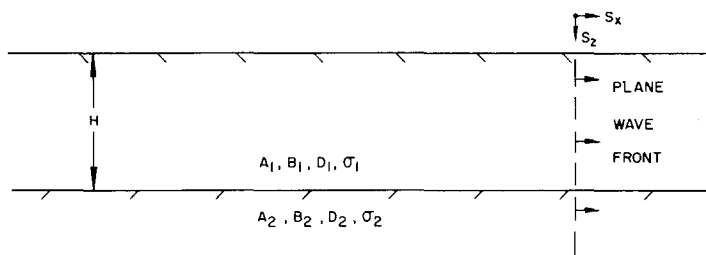


FIG. 1. Structural model and nomenclature.

are used:

- A = compressional velocity, alpha
- B = shear velocity, beta
- D = density
- σ = Poisson's ratio, sigma

We have chosen to characterize a particular structural model by specifying the four quantities:

$$\sigma_1, \sigma_2, B_2/B_1, \text{ and } D_2/D_1.$$

Depth H can be absorbed into a dimensionless independent variable proportional to period, B_1T/H . The dependent variables proportional to phase and group velocity are taken as C/B_1 and U/B_1 .

A third dependent variable was also computed, the amplitude ratio S_x/S_z . This represents the relative magnitude of horizontal to vertical ground displacement at the free surface. It is an observable quantity and sometimes, in experimental work, a diagnostic one. A multiplying factor of $-i$ is understood for all ratios; the factor corresponds to retrograde motion and a 90° phase shift between the two components.

Alternative parameters to σ_1 and σ_2 are ratios involving the compressional velocities, for example, A_1/B_1 and A_2/B_2 . The two pairs are interchangeable through the

relation

$$(A/B)^2 = (2 - 2\sigma)/(1 - 2\sigma); \quad (1)$$

the differences in dispersion appear easier to understand using Poisson's ratios.

Results presented here are selected from a larger publication (Mooney and Bolt, 1965) which includes numerical values for all cases presented in this paper. Solutions were obtained for variations of each parameter of the structure about the following two basic models:

	<i>Model 1</i>	<i>Model 2</i>
σ_1	0.25	0.25
σ_2	0.25	0.25
B_2/B_1	1.5	2.0
D_2/D_1	1.3	1.5

These values were selected because they arise in certain physical applications, and because some of the more interesting and novel results begin to appear as B_2/B_1 is increased from 1.5 to 2.0.

For each of the two models, results were obtained for eight values of each of the parameters in turn, holding the other three fixed. The eight values are:

$$\sigma_1, \sigma_2 = 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45.$$

$$B_2/B_1 = 1.1, 1.2, 1.3, 1.4, 1.5, 2.0, 3.0, 4.0.$$

$$D_2/D_1 = 0.8, 1.0, 1.2, 1.3, 1.4, 1.5, 2.0, 3.0$$

The eight values of σ_1 and σ_2 correspond to

$$A_1/B_1 \text{ or } A_2/B_2 = 1.5000, 1.5584, 1.6330, 1.7321, 1.8708, 2.0817, 2.4495, 3.3166.$$

Results were calculated for the fundamental Rayleigh mode and for the first and second higher modes. These are sometimes identified with the M_{11} , M_{21} , and M_{12} modes, respectively (see Tolstoy and Usdin, 1953). The M_1 wave type of Sezawa has M_{11} and M_{12} as its first two modes; the M_2 type has M_{21} and M_{22} . Rayleigh modes beyond the fundamental have been called "shear" modes by a number of authors (see e.g. Oliver and Ewing, 1957, p. 201). This description arises from the property of the higher modes, not possessed by the fundamental mode, that the phase velocity approaches the shear wave velocity in the substratum at the long wavelengths.

THEORY

The Rayleigh wave period equation is derived from the equations of motion and the boundary conditions, and relates the velocity of propagation to allowable periods. For a single layer overlying a half space, the usual period equation is obtained by setting a 6×6 determinant equal to zero (Sezawa, 1927).

For the single surface layer, it is possible to reduce the period equation to the form

$$\Delta(C, k) = 0, \quad (2)$$

where Δ is a 3×3 determinant, given by Bolt and Butcher (1960), and k is the wave number. Real solutions of this equation exist under the condition that

$$1 \leq (C/B_1) \leq B_2/B_1 \quad (3)$$

which entails that

$$B_1 < B_2. \quad (4)$$

When $C = B_1$ or B_2 there are limiting cases but no computational difficulty arises at these points. For all modes except the fundamental, the upper end point ($C = B_2$) represents the long-period termination or cutoff of unattenuated propagation. (The continuation represents leaking mode propagation.) No restriction applies to A_1 or A_2 beyond the physical requirement that Poisson's ratio in both media must lie in the range 0.0 to 0.5; in particular, real solutions exist for either

$$A_1 < B_2 \quad \text{or} \quad A_1 > B_2. \quad (5)$$

We do not consider in this paper complex solutions to the period equation, corresponding to attenuated or leaking modes.

Group velocities have often been computed (e.g. Bolt and Butcher, 1960) by taking finite differences of the phase velocity values for substitution in the relation

$$U = C + k \, dC/dk. \quad (6)$$

The numerical differentiation may lead to significant errors, particularly in regions of rapid change in phase velocity, near the higher-mode cutoffs, and near the beginning and end of the range of computation.

Alternative methods have been devised for computing more accurate values of group velocity, for example using Rayleigh's principle (Meissner, 1926) and using energy density (Rosenbaum, 1964). For the problem considered here, however, the differentiation may be carried out analytically at the outset.

To obtain a value for dC/dk in equation (6), differentiation of equation (2) gives

$$\left(\frac{\partial \Delta}{\partial C}\right) dC + \left(\frac{\partial \Delta}{\partial k}\right) dk = 0. \quad (7)$$

Substitution into equation (6) then yields:

$$U = C - k \left(\frac{\partial \Delta}{\partial k} \right)_{\frac{\partial \Delta}{\partial C}}. \quad (8)$$

The partial derivatives are obtainable by differentiating successive columns in the determinant Δ . The same derivatives serve other purposes in the computation process as shown below.

To obtain a solution for equation (2), we choose a value for T , estimate a trial value for C (and therefore for k), and evaluate the determinant Δ together with its partial derivatives. An improved estimate for C can then be obtained by the second-

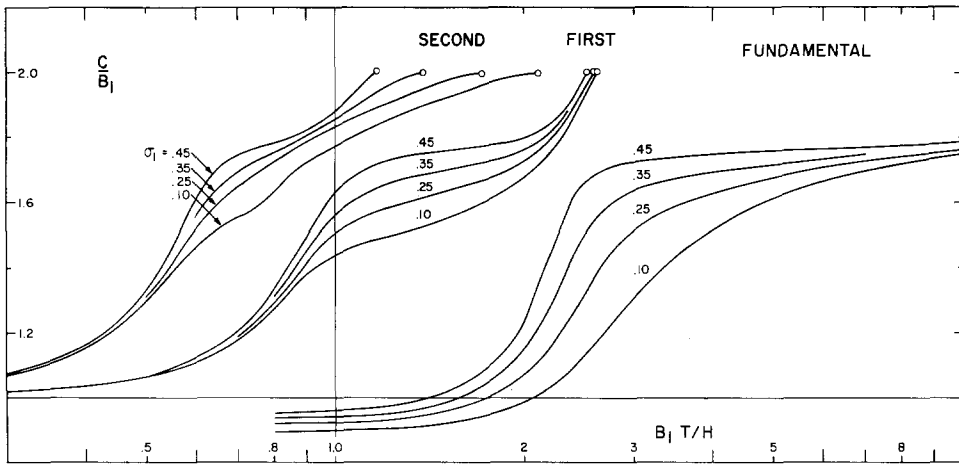


FIG. 2. Phase velocity versus σ_1 .

order Newton-Raphson iteration method given by

$$C_{n+1} = C_n - \Delta / \left(\frac{\partial \Delta}{\partial C} \right)_n. \quad (9)$$

This process was repeated until successive values differed by less than 10^{-5} .

On automatically incrementing T by ΔT , the partial derivatives can be used to estimate a new trial value for C . The evaluation of cutoff points for the higher modes was carried out as a separate step, by solving equation (2) for k (hence T) subject to the condition that $C = B_2$. Amplitude ratios were obtained from the boundary conditions by substitution of the solutions of equation (2).

The tabulated results were obtained using computer arithmetic which works effectively to 10 decimal places. Numerical checks were performed in several ways. A portion of the tables was checked using double-precision Fortran. For some models, comparative values were computed using a program made available by

Dr. James Dorman which uses single-precision Fortran and a different computational scheme. A comparison was also made with the tables given in Bolt and Butcher (1960). As far as we can determine, all tabular values are correct to the last decimal place listed.

DISCUSSION OF RESULTS

General. A selection from the computed results is presented in Figures 2-17. Unless otherwise indicated, parameters are $\sigma_1 = 0.25$, $\sigma_2 = 0.25$, $B_2/B_1 = 2.0$, $D_2/D_1 = 1.5$. Small circles which terminate curves represent long-period cutoff points. Curves terminated without circles have been shown incompletely for clarity of presentation.

An important general conclusion relates to the period ranges over which significant variations are observed. In terms of the dimensionless quantity B_1T/H

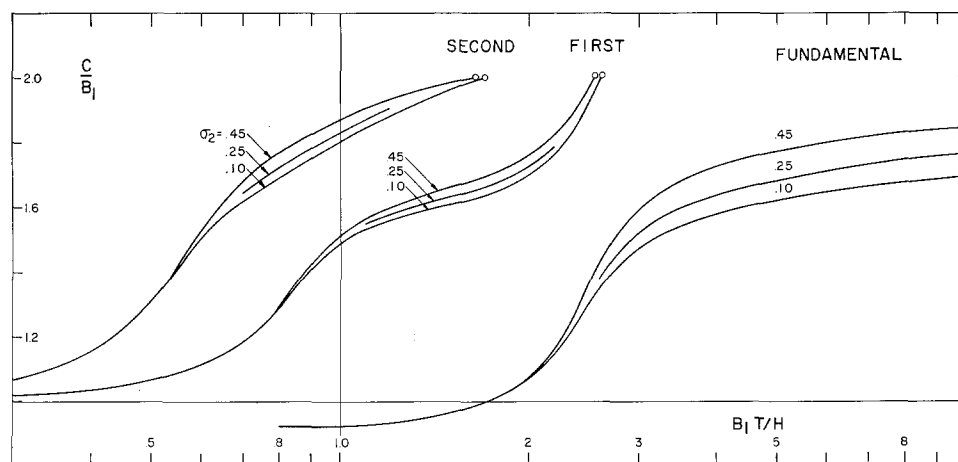


FIG. 3. Phase velocity versus sigma 2.

which is proportional to period, these ranges are largely independent of the parameters of the structure (except as they influence cutoff):

Fundamental mode	$B_1T/H = 1 \text{ to } 20$
First higher mode	$= 0.3 \text{ to cutoff}$
Second higher mode	$= 0.2 \text{ to cutoff.}$

Another general conclusion relates to the relative importance of the parameters. As is known from other studies the shear velocity ratio, B_2/B_1 , exerts the most important influence on Rayleigh wave dispersion for all modes. The present calculations show that Poisson's ratio in the surface layer, σ_1 , and the density ratio, D_2/D_1 , produce substantial but lesser effects. Poisson's ratio in the half space, σ_2 , is least significant. From equation (1), it may be stated also that compressional

velocity in the surface layer produces a significant effect on the dispersion results, but that compressional velocity in the half space is less important.

For the fundamental mode, both group and phase velocity curves approach the characteristic Rayleigh wave velocity for a uniform half space with properties of

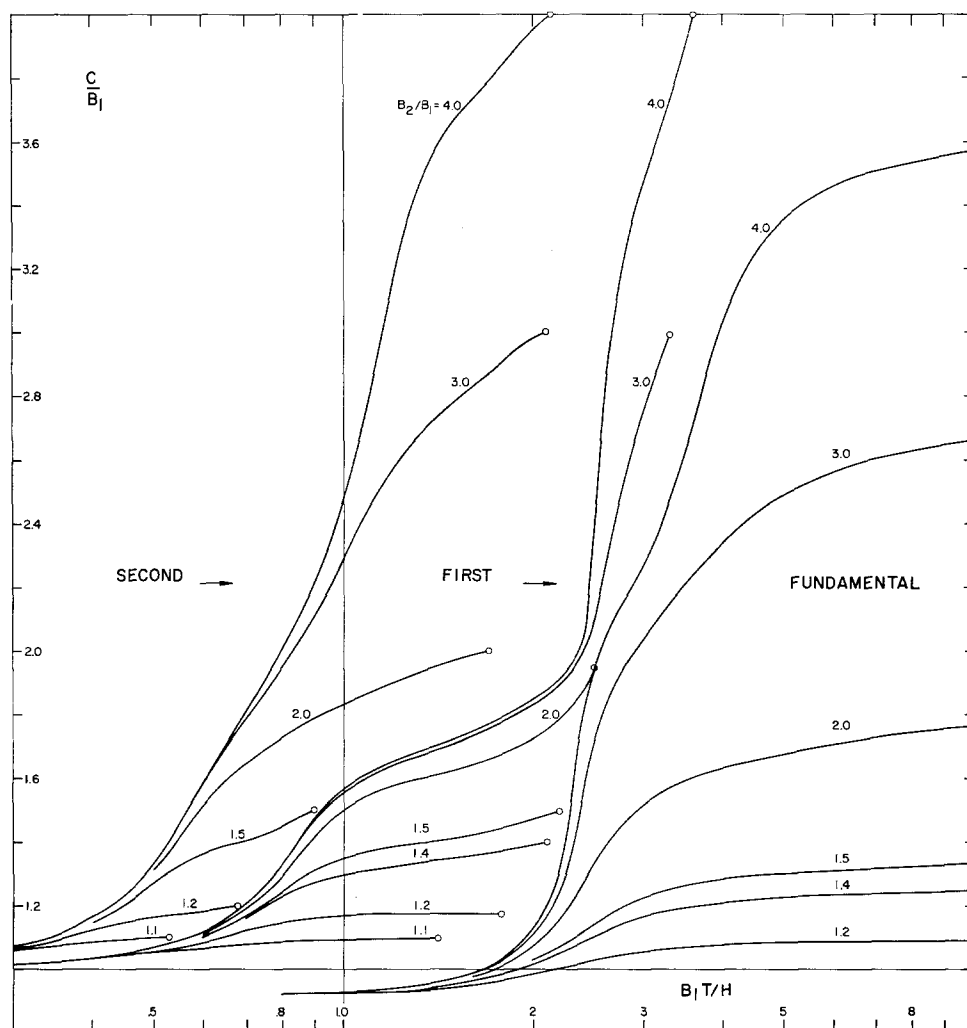


FIG. 4. Phase velocity versus shear velocity ratio.

the layer in the short-period limit; in the long-period limit, the curves approach the Rayleigh velocity for a uniform half space identical to the substratum. For all higher modes, group and phase velocities approach the shear velocity in the layer as the periods decrease.

Phase Velocities. The principal effect on phase velocity due to σ_1 (Figure 2) is to steepen the curve near its inflection point. Variations due to σ_1 are largely confined

to the ranges:

Fundamental mode $B_1 T/H = 2$ to 5

First higher mode $= 0.9$ to 2

Second higher mode $= 0.6$ to 0.9 .

Figure 3 shows that the effects due to variations in σ_2 are less significant and appear at somewhat longer periods.

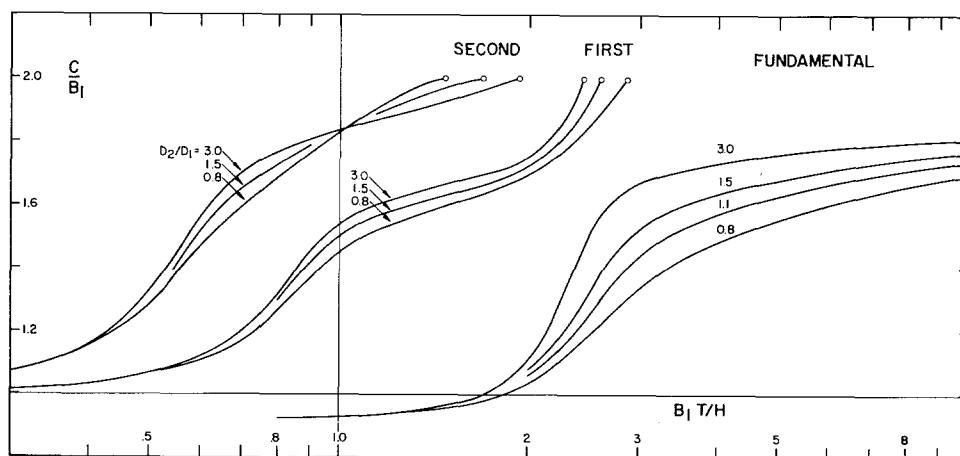


FIG. 5. Phase velocity versus density ratio.

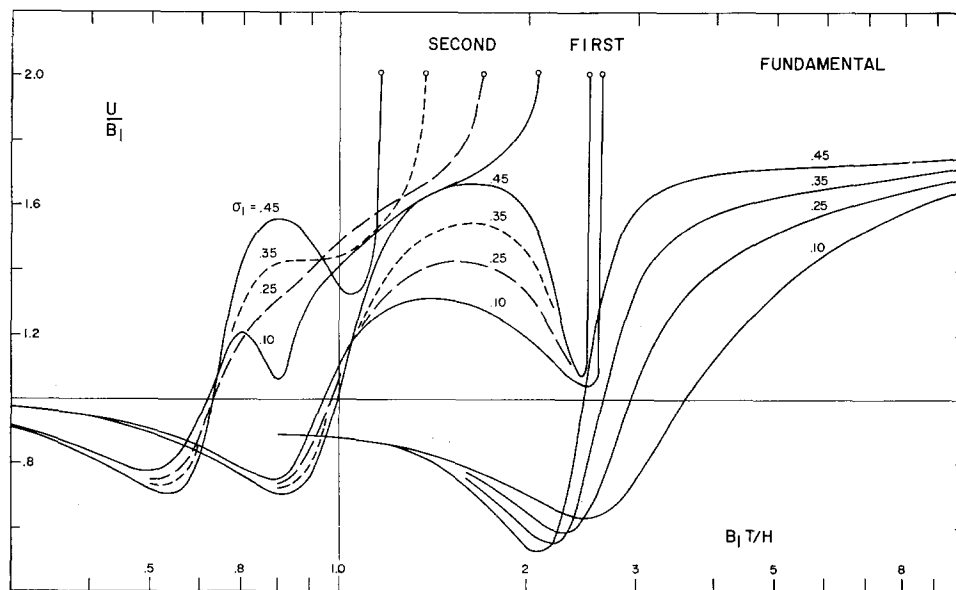


FIG. 6. Group velocity versus σ_1 .

Shear velocity variations produce major changes in phase velocities (Figure 4) for all modes, although the differences disappear toward the short period end. The first higher Rayleigh mode curves show a tendency toward a horizontal portion at the longer periods just before cutoff. Both first and second higher modes show a saturation effect at the shorter and medium periods, such that a large increase in B_2/B_1 produces a relatively small change in phase velocity.

Increased density contrast increases the phase velocity of waves of the fundamental mode (Figure 5). The changes are smaller in the case of the first and second higher modes; the second higher Rayleigh mode curves converge toward a common point near $B_1T/H = 1.0$ for all values of the density ratio.

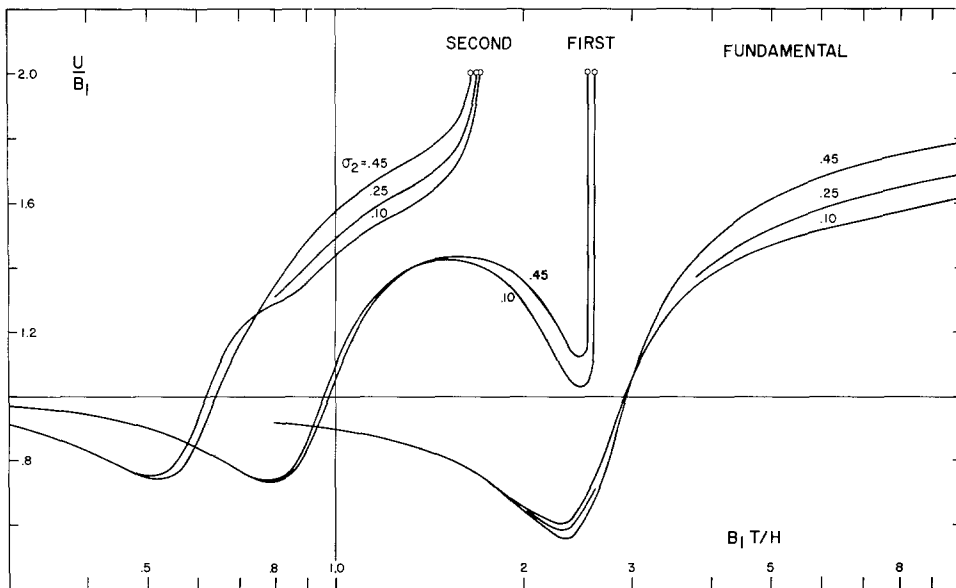


FIG. 7. Group velocity versus σ_2 .

Group Velocities. A prominent feature on all group velocity curves (Figures 6–10) is the normal short-period group velocity minimum. As the mode number increases, this minimum falls at a shorter period which is roughly independent of the parameters of the structure, except that the smallest shear velocity ratios displace the minimum toward smaller periods. Typical positions for the principal group velocity minima for the fundamental and for the first and second higher Rayleigh modes are $B_1T/H = 2.3, 0.8$, and 0.5 .

In addition to the normal short-period minimum, one or more group velocity minima appear at the longer periods in some models. These stationary points would produce narrow-band signals on seismograms; observed pulses such as the phase *Sa* have been attributed to such features (e.g. Bolt and Dorman, 1961; Anderson and Toksöz, 1963). They occur on the fundamental mode only for very large shear velocity ratios, but seem to be a characteristic feature of the higher modes. As many as three group-velocity minima may be seen on some of the second higher mode

curves, and a fourth appears for B_2/B_1 greater than 4.0. It should be noted (Figure 6 and Figure 8) that the ranges

$$0.2 < \sigma_1 < 0.3 \quad \text{and} \quad 2 < B_2/B_1 < 3$$

seem to represent a special combination for the second higher mode, in that all additional group velocity minima disappear.

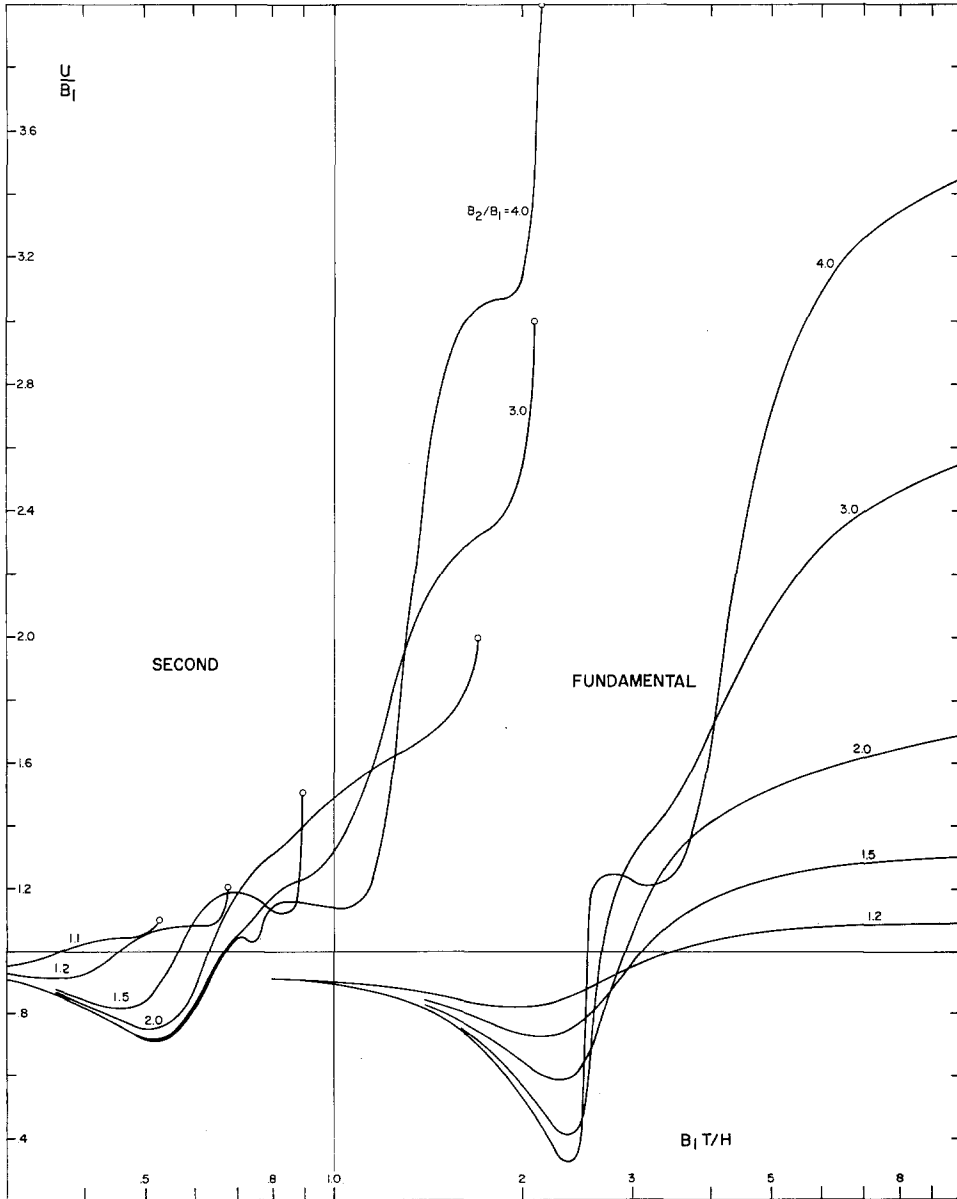


FIG. 8. Group velocity versus shear velocity ratio (1).

A sharp group-velocity minimum appears on the first higher Rayleigh mode just preceding cutoff. The arrival time for this signal will depend strongly upon the shear velocity ratio, arriving earliest for the intermediate value of $B_2/B_1 = 1.5$. A third minimum develops when B_2/B_1 exceeds 3.0.

The first higher Rayleigh mode exhibits a broad group-velocity maximum near

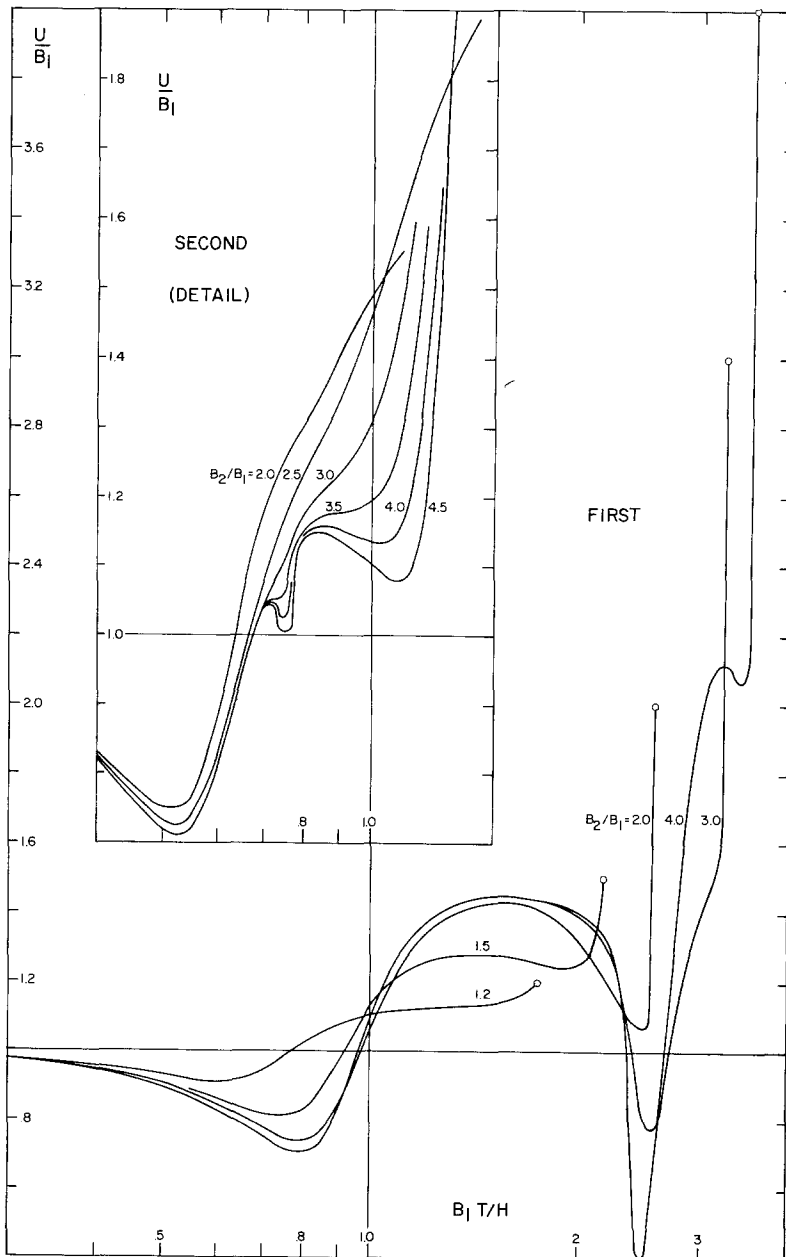


FIG. 9. Group velocity versus shear velocity ratio (2).

$B_1T/H = 1.5$. Such a maximum would correspond to an early pulse-like arrival with moderately large amplitudes. The dependence upon σ_1 of the arrival time for these pulses (Figure 6) is particularly striking with large σ_1 corresponding to earlier arrivals.

The effects of density variations are relatively small for all three modes (Figure 10) and are largely confined to the long-period portions.

Surface Displacements. A physical interpretation of the surface displacement curves (Figures 12–16) is sketched in Figure 11. Positive values of the ratio, S_x/S_z , correspond to retrograde particle motion and negative values to progressive or prograde motion. Points where the curve crosses the horizontal axis yield purely

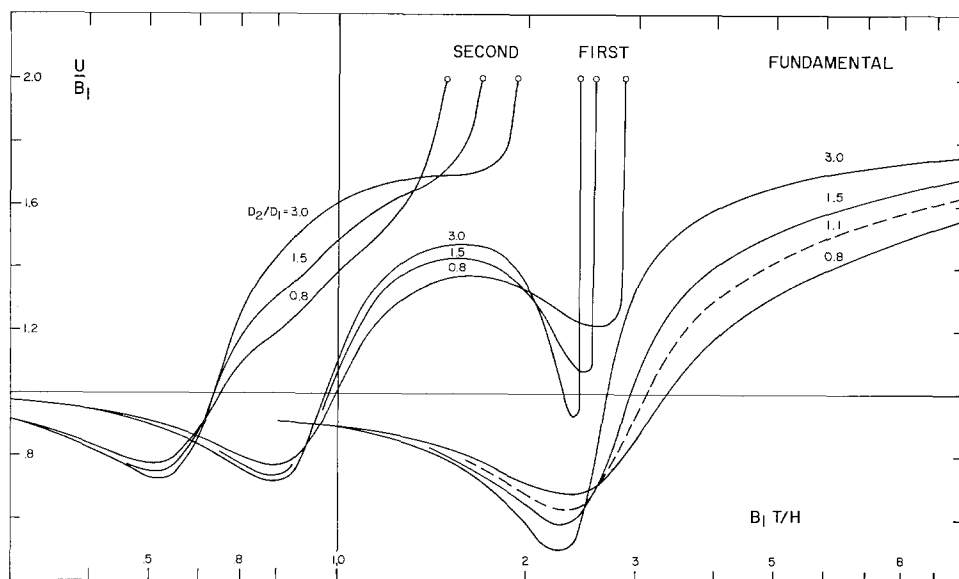


FIG. 10. Group velocity versus density ratio.

vertical surface displacement. Points where the curve approaches infinity yield purely horizontal displacement.

The displacement ratio depends most strongly upon the shear velocity ratio (Figure 15) but strongly also upon σ_1 (Figures 12, 13). The dependence upon density ratio (Figure 16) and σ_2 (Figure 14) is much less important.

For shorter periods, all three modes have normal retrograde motion, with S_x/S_z in the range 0.5 to 0.7. The asymptotic value decreases by roughly 0.05 for each successively higher mode.

All motion in the fundamental mode is retrograde for the model parameters used here, although an example involving layered alluvium will be given later to show that this is not always true. The horizontal component increases at the larger periods, reaching a maximum near $B_1T/H = 4$. The ratio S_x/S_z remains less than 1.5 for all except the largest shear velocity ratios, where values up to 10 are encountered.

Except for the smallest shear velocity ratios, less than 1.4, the first and second higher modes show prograde surface motion for the longer periods. The long period motion for the second higher mode becomes dominantly horizontal for nearly all values of the structural parameters except small shear velocity ratios.

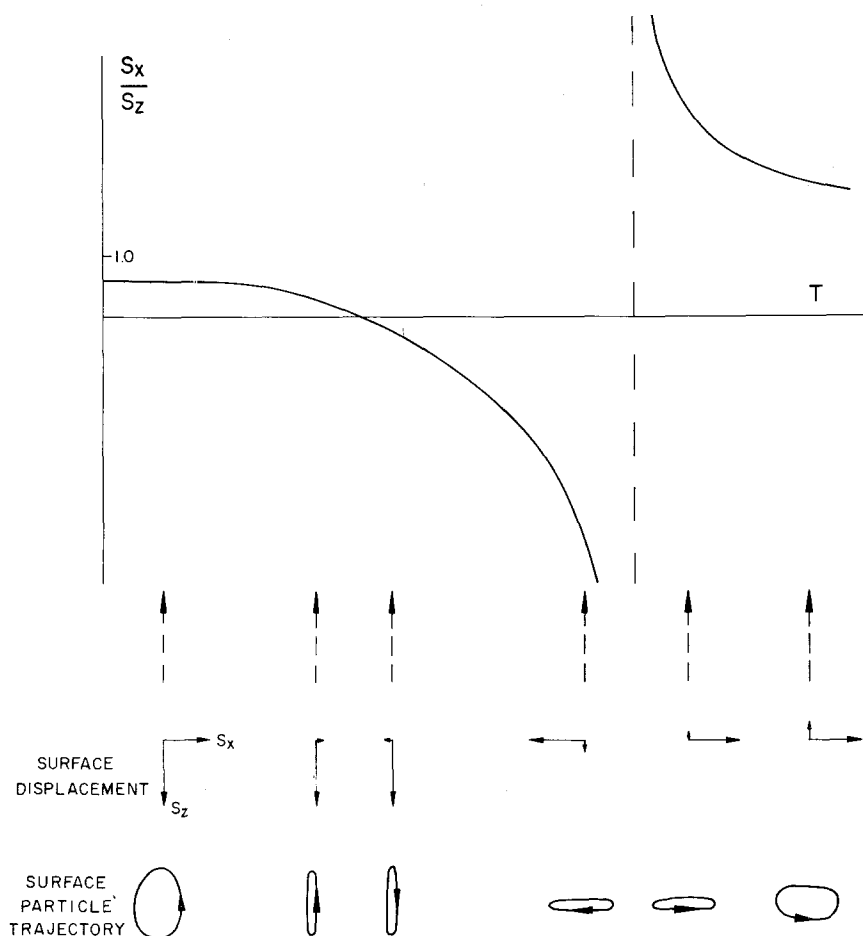


FIG. 11. Physical interpretation of a typical amplitude ratio curve.

The first and second higher modes pass through regions where the surface motion is completely vertical, near $B_1T/H = 0.8$ and 0.5 respectively, for all parameters except small shear velocity ratios.

Long-Period Cutoffs. The long-period cutoff values (Figure 17) are also more sensitive to the shear velocity ratio, B_2/B_1 , than to the other parameters. The dependence is most pronounced for ratios less than 2.2. Above this value, the second shear mode cutoff becomes nearly constant.

The effect of variations in σ_1 is insignificant except for the second higher Rayleigh mode with shear velocity ratios greater than 1.5. The effect of σ_2 is negligible. Den-

sity variations produce appreciable changes in the cutoff values, particularly for the larger shear velocity ratios.¹

GEOPHYSICAL EXAMPLES

In this section we illustrate the preceding results by means of specific geophysical and laboratory examples. These examples emphasize some of the dispersion properties discussed earlier and suggest the wide range of problems for which the new tables may be used. The specific parameters selected are tabulated in the Appendix, together with source references.

It should be emphasized that the conclusions reached in the following sections remain valid for layer thicknesses other than those selected. The only effect of increasing the layer thickness by a factor is to multiply the period scale (horizontal

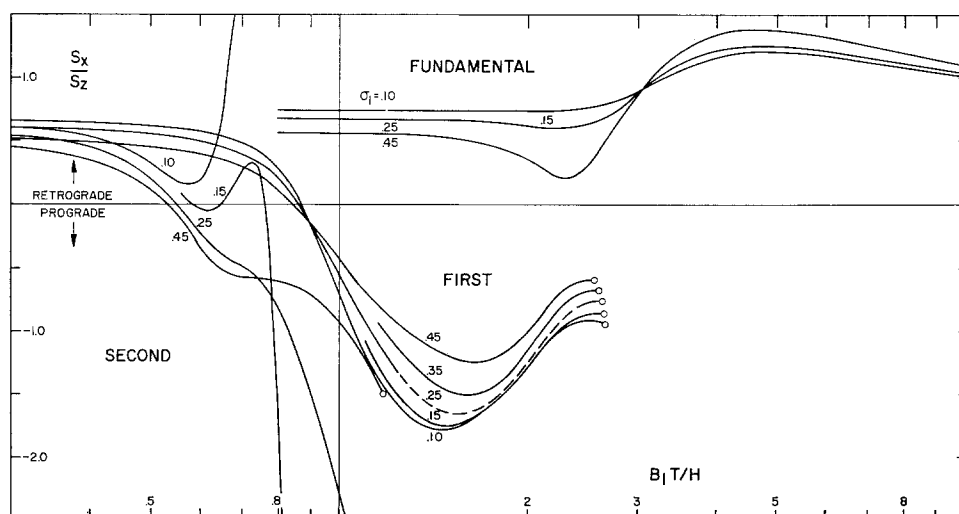


FIG. 12. Amplitude ratio versus σ_1 (1).

axis) by the same factor. To the extent to which the single-layer model is a valid approximation in a particular application, this characteristic of the curves provides a means for deducing layer thickness from observational data.

Continental Ice Cap. Consideration has already been given to surface wave dispersion as a means to estimate thickness of the Antarctic ice cap (Dewart and Toksöz, 1964). The present calculations show that the higher Rayleigh modes can be more sensitive to ice thickness than the fundamental mode.

If we assume normal upper crustal material beneath the ice, then the problem is characterized by a moderately large shear velocity ratio, 1.8, a large density ratio, 2.9, and a relatively high Poisson's ratio, 0.33, in the ice layer. Figure 18 shows the results which may be expected.

¹ A reviewer has kindly offered an interpretation for the first graph in Figure 17. He suggests that cutoff for the first shear mode is controlled entirely by the shear velocity in the layer, whereas that for the second shear mode depends largely upon the compressional velocity in the layer.

A fundamental mode Airy phase should arrive very late on the record, with a period of 2.6 seconds for the assumed ice thickness of 2.3 km. The period T at which the Airy phase actually occurs may be used to deduce ice thickness H from the relationship $T/2.6 = H/2.3$.

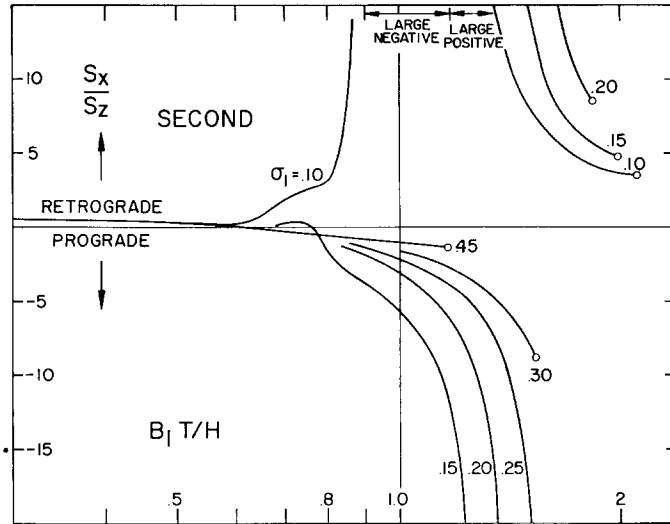


FIG. 13. Amplitude ratio versus sigma 1 (2).

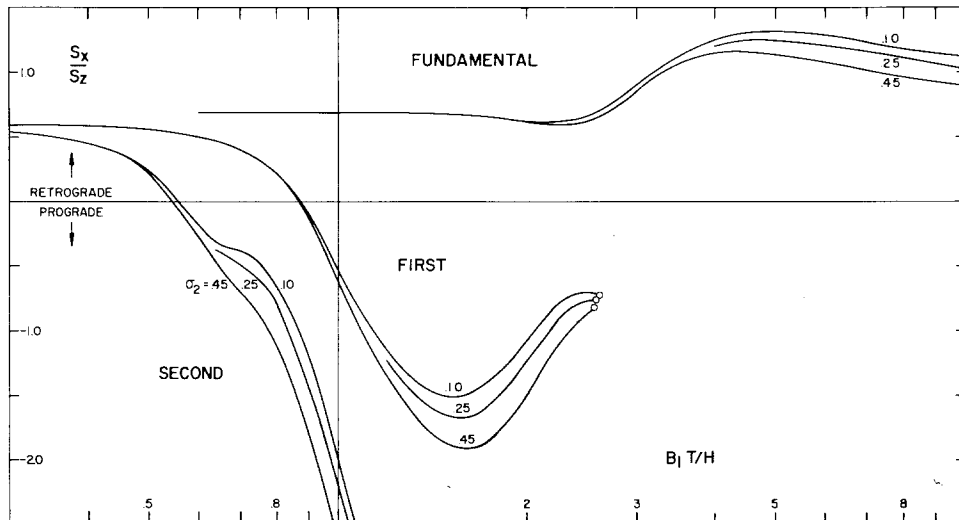


FIG. 14. Amplitude ratio versus sigma 2.

The first and second higher Rayleigh modes each show two group velocity minima. The more prominent minima occur at periods of 0.96 and 0.62 seconds respectively. On seismograms, the effect would be the nearly simultaneous arrival of two pulses just preceding the fundamental mode Airy phase.

The first higher mode has a broad group velocity maximum. Again, the effect

on a seismogram would be a pulse-like signal with periods centered around 1.9 seconds and an arrival time corresponding to a velocity of about 2.95 km/sec.

All three group velocity curves show steeply rising limbs to the right of the group velocity minima. These portions of the curves would be associated with extended constant-frequency wave trains such as those associated with oceanic paths. It

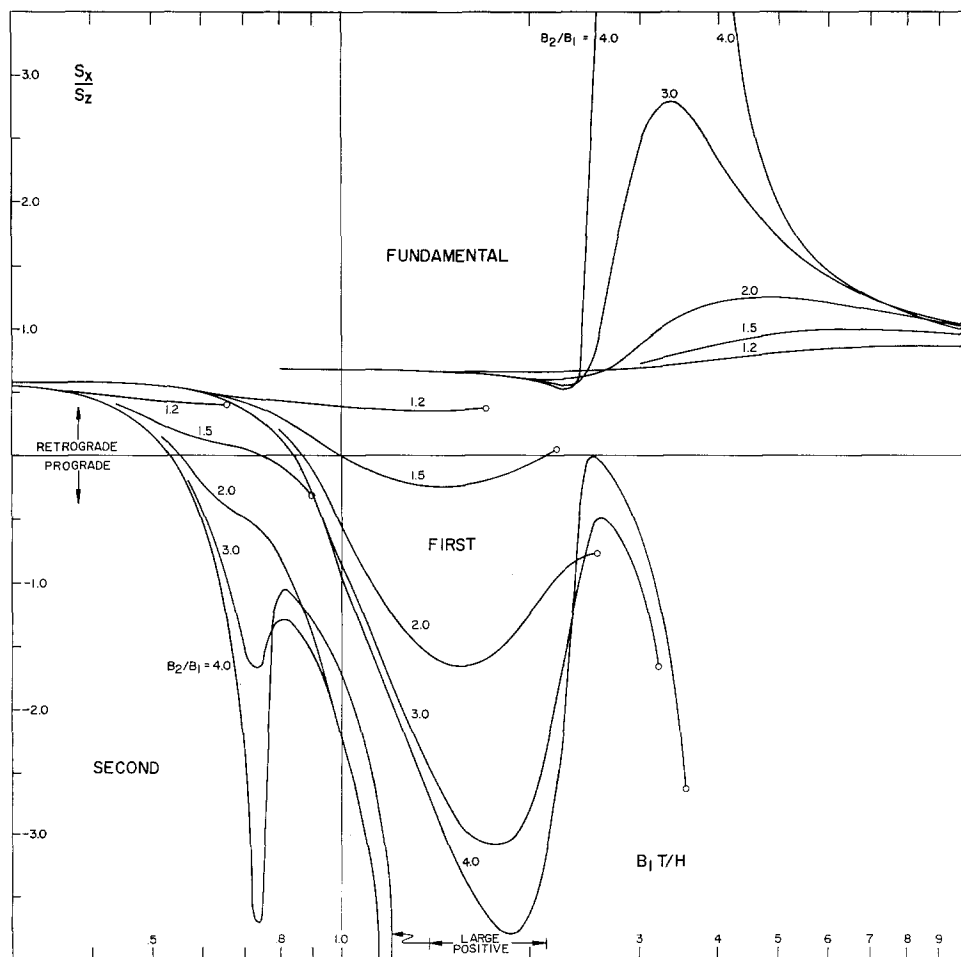


FIG. 15. Amplitude ratio versus shear velocity ratio.

might be expected, therefore, that measurement of the dominant frequency would provide a sensitive means for determining ice thickness.

The curves for surface displacement in Figure 18 permit several predictions: (1) The fundamental mode should show a large horizontal displacement component at the longer periods. (2) The first and second higher Rayleigh modes should produce purely vertical displacement at periods near 1.0 and 0.65 seconds respectively, with prograde motion for all larger periods. (3) The long period motion for both higher modes should have a strong horizontal component.

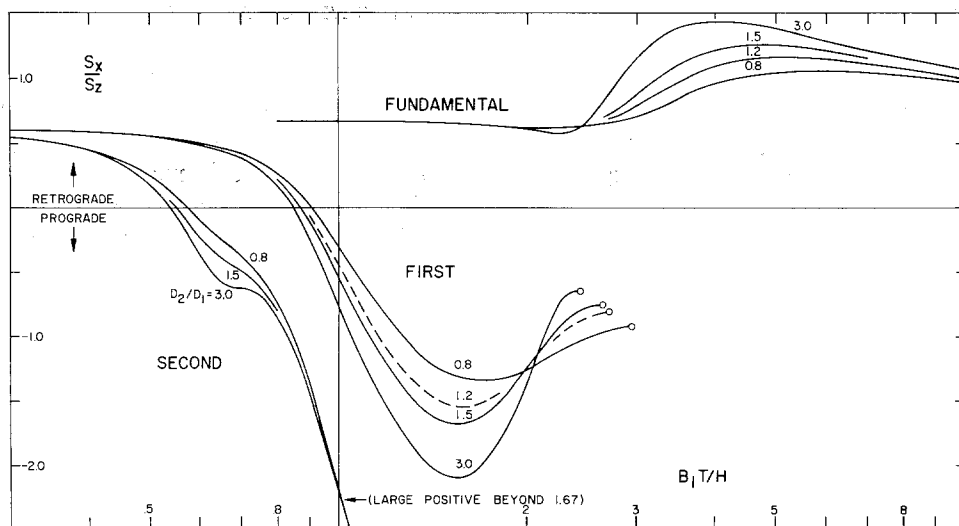


FIG. 16. Amplitude ratio versus density ratio.

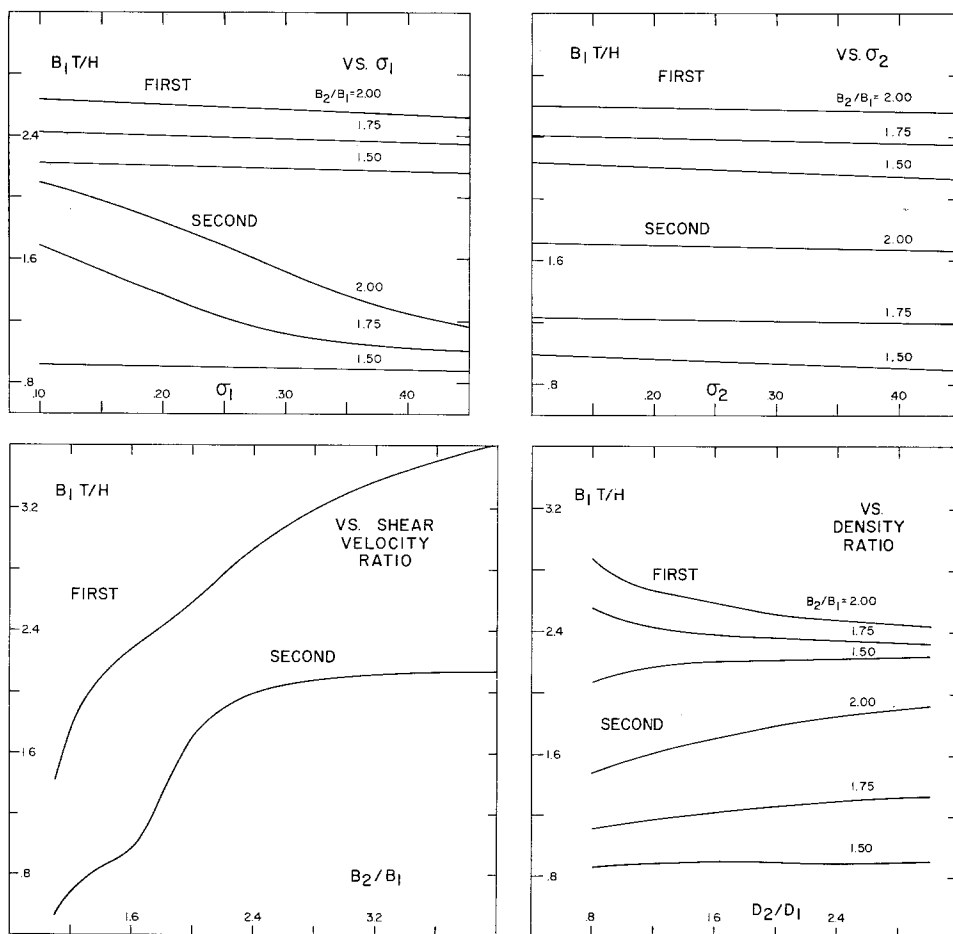


FIG. 17. Dependence of long-period cutoff on structure parameters.

As noted earlier, the shape of these curves is independent of ice thickness. Inspection of Figure 16 shows that density variations in the ice will not produce major effects on the amplitude curves, particularly within the narrow range of uncertainty for mean ice density. The dependence upon shear velocity ratio may be seen in Figure 15. We conclude that observations of the surface displacements at various periods, especially for the higher modes, should lead to information on the shear velocity ratio in the ice.

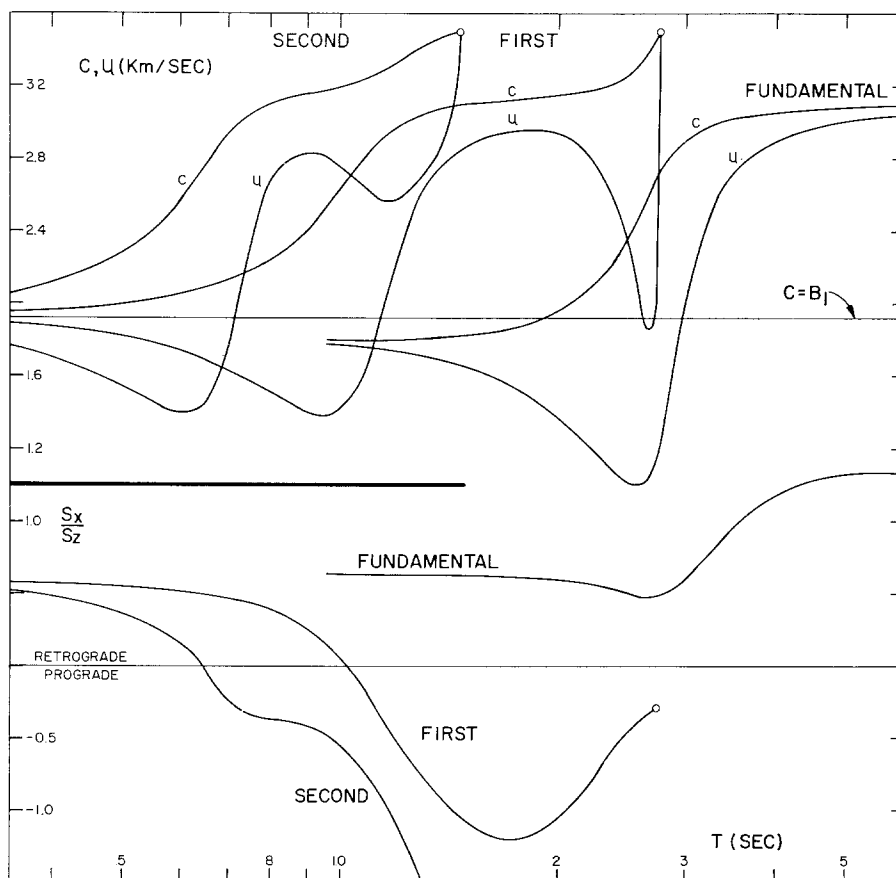


FIG. 18. Dispersion results for continental ice cap, thickness 2.3 km.

Continental Crust. A single-layer model with plane boundaries cannot be expected to represent adequately all properties which arise from the structural complexity now accepted as existing in the crust and upper mantle. Nevertheless, several interesting conclusions are suggested by a single-layer approximation, in line with Dorman's (1959) suggestion that "the weighted mean crustal velocity is a valuable parameter in the interpretation of dispersion data".

Figure 19 shows the behaviour of phase velocities, group velocities, and amplitudes for a standard continental crust (Press *et al.*, 1956). The group velocity minima

for the first three Rayleigh modes occur at 21.5, 7.0, and 4.0 seconds. No group-velocity maxima appear, although the second higher Rayleigh mode curve levels off just preceding cutoff. The steeper slopes for the higher modes would support the suggestion of Oliver and Ewing (1957) that they provide a better measure of crustal thickness than the fundamental mode.

The amplitude ratios are retrograde for all modes and for all periods. This special property, noted earlier by Oliver and Ewing (1957), arises from the small shear velocity contrast as may be seen in Figure 15. The fundamental mode has nearly equal horizontal and vertical displacements for all periods, but the higher modes have predominantly vertical displacements at the longer periods.

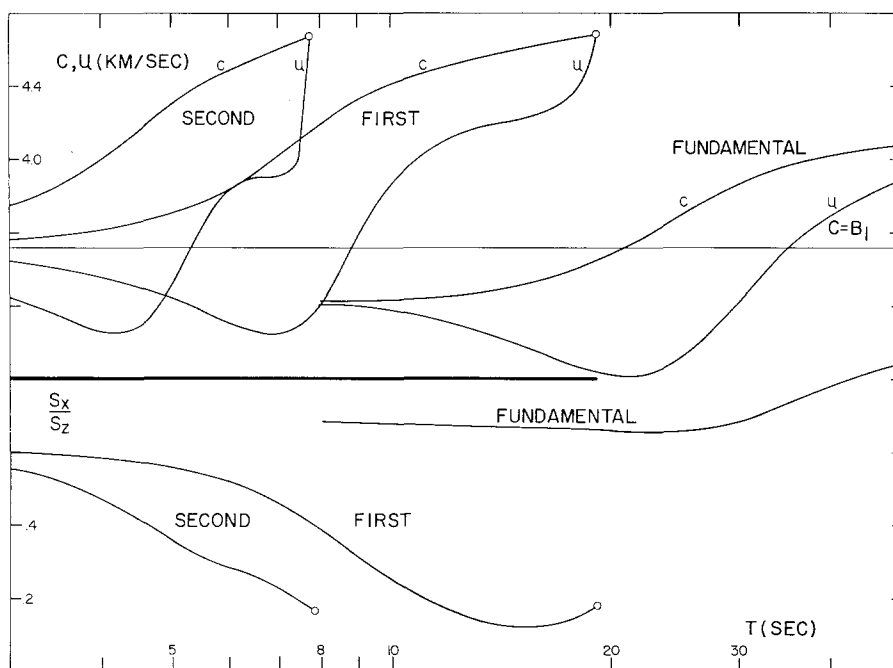


FIG. 19. Dispersion results for standard continental crust, thickness 35 km.

Sedimentary Basin. Surface wave dispersion across deep sedimentary basins offers a means for mapping thickness variations and deducing averaged properties of the sedimentary column (see Dobrin *et al.*, 1954; McEvelly and Stauder, 1965). Figure 20 suggests some of the effects which might be expected for a section of 3 kilometers thickness. This model is characterized by negligible density contrast with moderately large value of σ_1 and shear velocity ratio.

We note first that the observations must extend to periods of 8–10 seconds in order to define the properties of the fundamental mode. The important data for the higher modes occur between 0.8 and 4 seconds.

Group velocity minima appear at 4.4, 1.6, and 1 seconds for the fundamental and the first and second higher modes. The group velocity is almost identical, 1.05 km/sec, hence their superposition may yield a larger signal than otherwise. The

first higher mode shows a broad group velocity maximum. This would yield a pulse-like arrival with a travel time corresponding to a velocity of 2.36 km/sec. Unlike the continental ice cap results, the second higher mode does not produce a group velocity maximum.

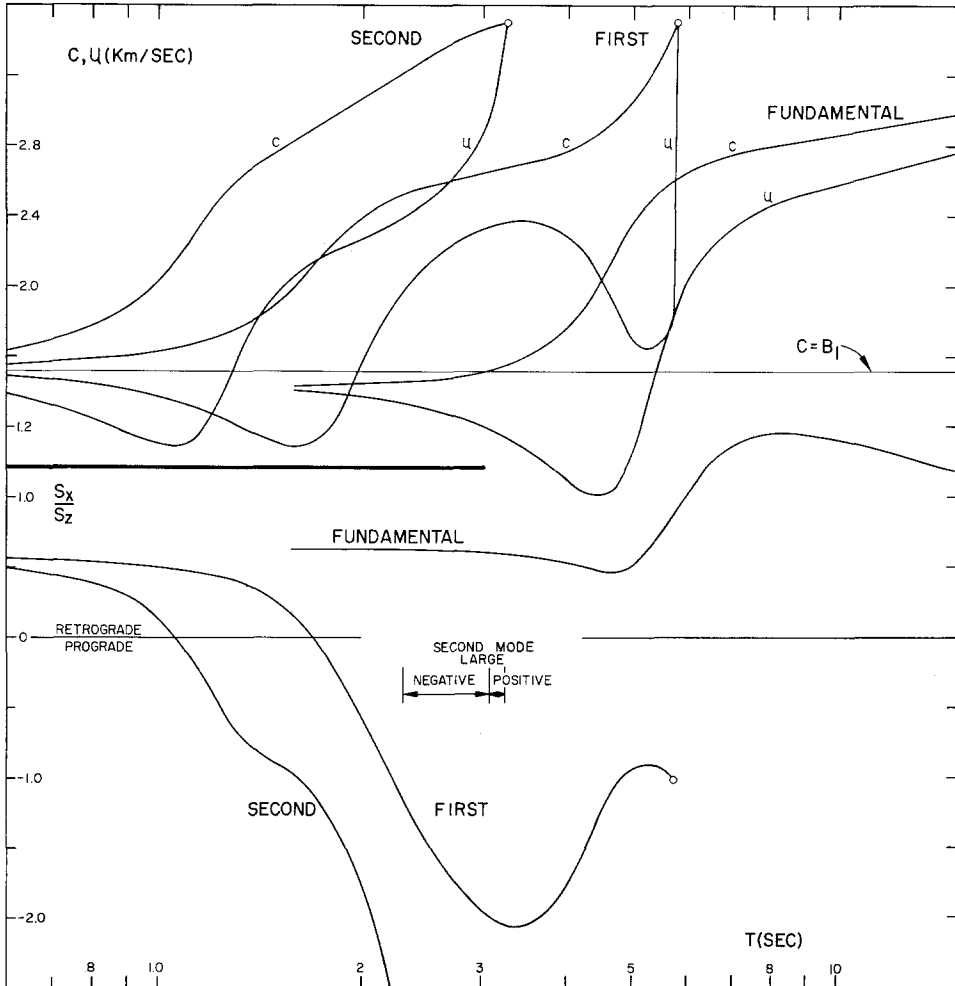


FIG. 20. Dispersion results for sedimentary basin, thickness 3 km.

The displacement ratios depend strongly upon period and should permit interpretation in terms of layer thickness or mean shear velocity.

Alluvium. An interesting application for surface wave dispersion methods is the determination of the properties of unconsolidated alluvium (Levshin, 1962). In this case, simple artificial sources can be used to provide data on alluvial thickness and physical properties averaged over large distances.

This problem involves a very large shear velocity ratio and an unusually large value for σ_1 (0.40 to 0.45). As a specific example for which precise shear velocity

data are available, we have chosen the underlying bedrock to be the Late Cretaceous Pierre Shale (McDonal *et al.*, 1958), in the Rocky Mountain area. The results for other choices would not be greatly different. Thickness is taken arbitrarily as 100 meters.

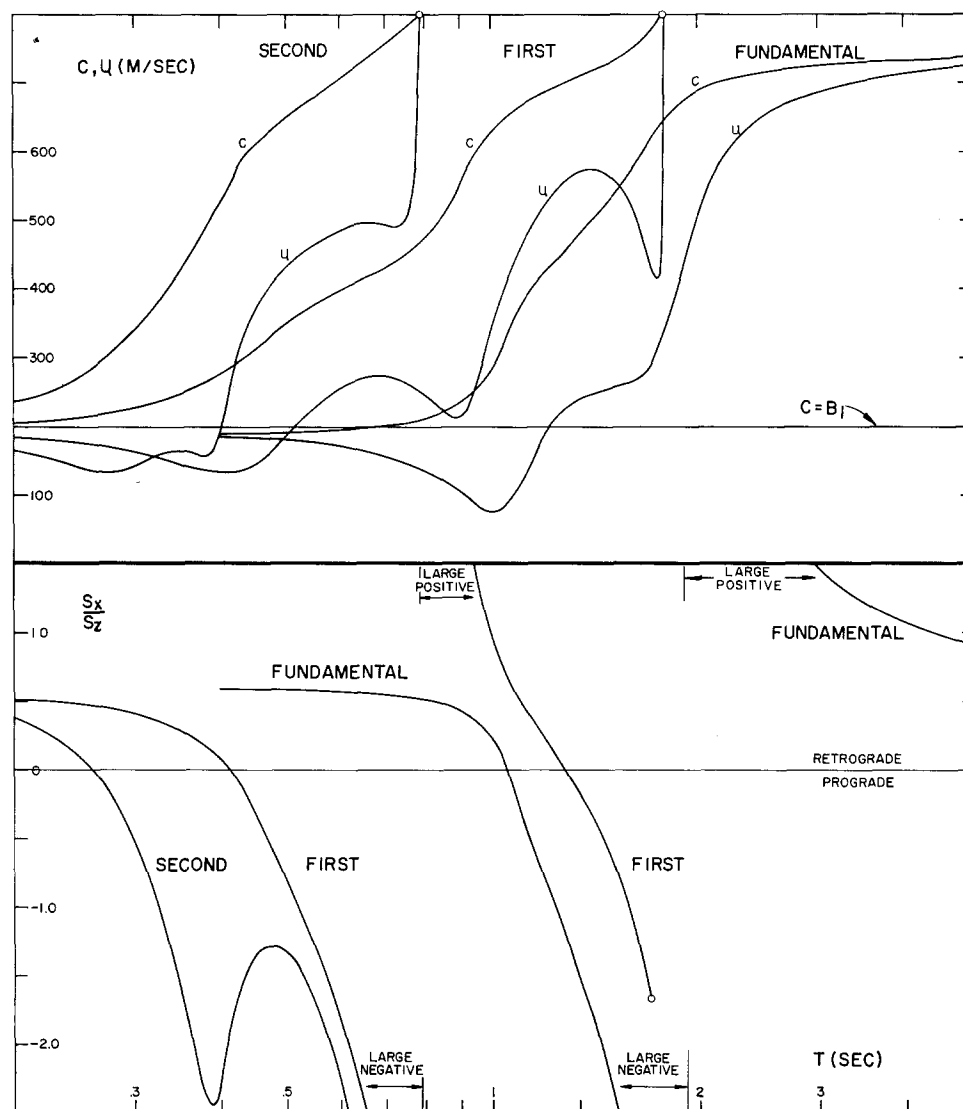


FIG. 21. Dispersion results for alluvium, thickness 100 m., over shale.

Figure 21 shows the complicated dispersion pattern to be expected. The fundamental mode group velocity exhibits two steeply rising limbs separated by an inflection. The first and second higher Rayleigh modes produce no less than three group-velocity minima each. The dominant group velocity minimum for all three

modes is unusually broad, and the group-velocity maximum for the first higher mode is unusually narrow. The Airy phase from the fundamental mode travels very slowly and will appear much later than the major part of the dispersed wave train.

It should be noted that observational data would be required at relatively long periods, at least up to 2 seconds for a 100 meter layer or 0.5 second for a 25 meter layer.

The combination of high shear velocity ratio and high Poisson's ratio yields the only example we have encountered of *prograde motion in the fundamental mode*. The prograde motion is confined to the narrow period range 1.05 to 1.90 seconds. This phenomenon may account for the prograde motion observed by Jones *et al.* (1963) for an alluvial overburden. If so, the period at which it occurs will depend very strongly upon thickness of the alluvial layer.

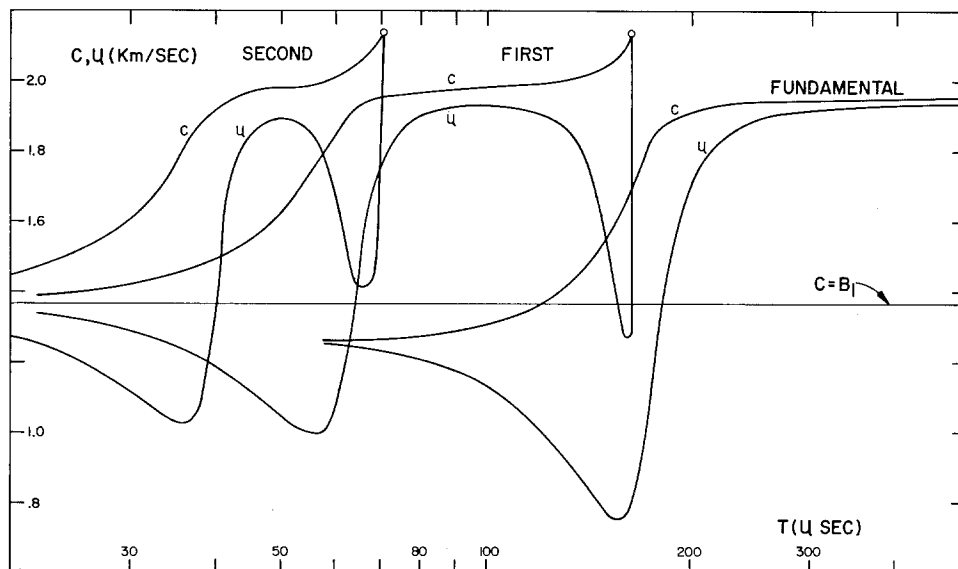


FIG. 22. Dispersion results for two-dimensional plexiglass-brass seismic model.

All of the modes pass very steeply through the region of dominantly vertical motion, hence we may conclude that most of the surface displacement will be horizontal except at the shorter periods.

Laboratory Seismic Models. As a supplement to the geophysical examples, dispersion curves were computed for a variety of laboratory seismic models. Model parameters were select to illustrate unusual features of the dispersion characteristics.

Space limitations preclude more than a single example. Data for the models are given in the Appendix. Tabulated values for these and other proposed laboratory seismic models are given by Mooney and Bolt (1965).

The effect produced by an extreme density contrast is shown in Figure 22. A large density in the half-space can be modelled by means of a layer of plexiglass overlying brass. Only the density ratio of 6.86 is extreme; the shear velocity ratio of 1.56 is moderate, as are the Poisson's ratios of 0.25 and 0.28.

As could be predicted from Figures 5 and 10, the group velocity curves for the first and second higher Rayleigh modes have each developed two pronounced minima separated by broad maxima. The pulses associated with the latter would arrive nearly simultaneously with each other and also with a similar pulse from the fundamental mode. As a result, we would expect at distance D considerable energy arriving at time $D/1.91$.

Final Example. We also include dispersion curves for a single-layer model treated earlier by Tolstoy and Usdin (1953). Their results have been reproduced elsewhere (Ewing *et al.*, 1957). The model had been selected by Jardetzky and Press (1952)

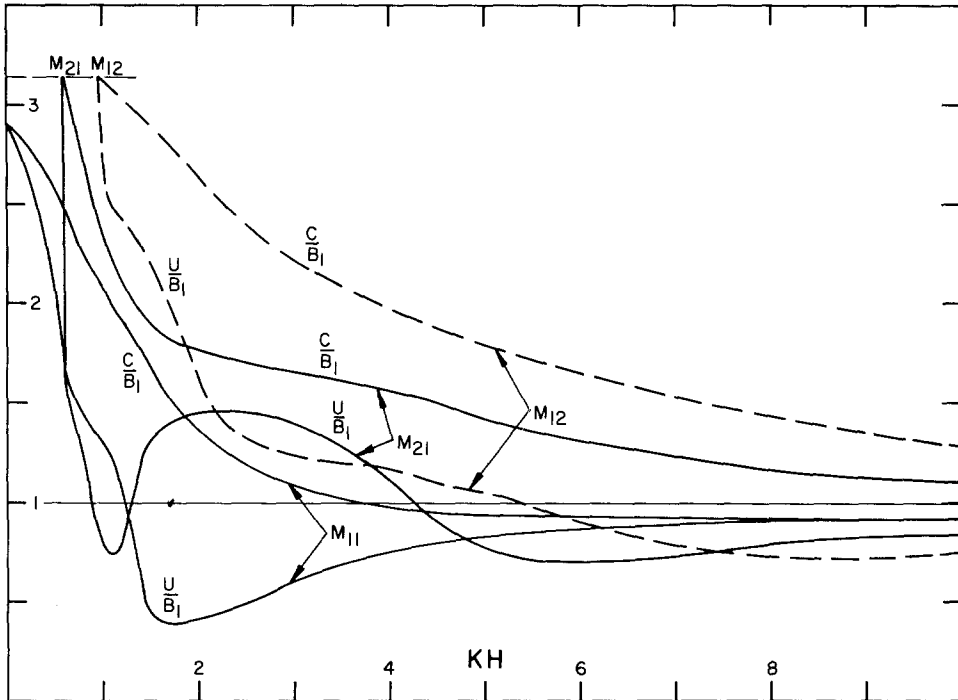


FIG. 23. Phase and group velocities for Tolstoy-Usdin single-layer model:
 $\sigma_1 = \sigma_2 = 0.25$, $B_2/B_1 = 3.147$, $\bar{D}_2/D_1 = 1.39$.

to represent near-surface conditions in seismic exploration. Graphed against B_1T/H , proportional to period, the results are essentially the same as shown in Figures 4, 8, 9, and 15 for shear velocity ratio 3.0. Graphed against kH , proportional to wave number, to permit comparison with the results of Tolstoy and Usdin, they are shown in Figure 23.

The curves contain considerably more fine structure than was noted by Tolstoy and Usdin. The group velocity minimum for the first higher Rayleigh mode propagates at a velocity of 0.73 rather than 1.3, and the group velocity maximum is broader. The behaviour of the higher modes near cutoff is altered.

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APPENDIX

DATA FOR EXAMPLES

(Figures in parentheses are not taken directly from reference cited. They are either computed from data supplied, or taken from another source.)

	Thickness (km.)	Longitudinal Velocity (km./sec)	Shear Velocity (km./sec)	Density or Density Ratio	Poisson's Ratio	Reference
Continental crust	35	(6.08) (8.11)	3.51 4.68	— 1.25	0.25 0.25	Press <i>et al.</i> 1956
Ice cap	2.3	3.820	1.925	0.918	(0.328)	Mellor, 1964
Sedimentary column	3.0	3.05	1.52	2.4	0.33	Various sources
Upper crust beneath ice cap or sedimentary column	—	5.85	3.50	2.67	(0.22)	Crary, 1963
Alluvium	(0.1)	(0.570)	0.200	1.70	0.43	Dobrin, 1951; Barkan, 1962
Bedrock beneath Allu- vium (Pierre shale)	—	2.160	0.800	(2.200)	(0.4±)	McDonal <i>et al.</i> , 1958
Brass, plate	—	(3.86)	2.135	8.34	0.28	Oliver <i>et al.</i> , 1954
Plexiglass, plate	—	2.36	1.37	1.22	0.25	Oliver <i>et al.</i> , 1954

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